

# Med Aziz Hraghi

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## PROFILE

Final-year engineering student specializing in Digitalisation & Data Analysis, focused on **AI engineering, machine learning, and intelligent systems**. Experienced in building production-oriented AI applications across **RAG, multi-agent systems, computer vision, and ML-powered decision support**, with industry and research experience delivering end-to-end AI systems.

## EXPERIENCE

**AI Engineer Intern** — *Talan Tunisie (Summer Camp 2026)*

Jul 2026 – Aug 2026 (On-site)

- Developed **Khatatli UrbanTwin**, an AI-powered urban-planning co-pilot transforming land parcels into regulation-aware, CAD-ready site plans — deployed live at [khatatli-urban-twin.vercel.app](https://khatatli-urban-twin.vercel.app).
- Engineered a **RAG pipeline** over **8 French legal codes** and local PLU zoning rules using Mistral embeddings and Qdrant, producing regulation answers with **verifiable legal citations**.
- Built a **LangGraph multi-agent pipeline** aggregating **20+ public data sources** across climate, elevation, soil, flood risk, mobility, cadastre, and POIs to automate terrain analysis and feasibility assessment.
- Developed generative planning features combining **2D site-plan generation, DXF/GeoJSON/PDF export, and 3D digital-twin visualization**, plus a 15-minute-city accessibility evaluator spanning **9 service categories**.
- Contributed within a **12-member engineering team**, implementing EU AI Act transparency mechanisms and writing **733 automated offline tests** across a FastAPI + React 19 / TanStack Start stack.

**Applied AI Intern** — *Institut National Agronomique de Tunisie (INAT) – LSTE*

Jul 2026 – Aug 2026

(Remote)

- Developed a **scientific-intelligence pipeline** ingesting **ArXiv, PubMed, and Scopus**, using embedding-based deduplication, semantic retrieval, and automated summarization to support research monitoring.
- Built an **AI-assisted irrigation decision-support module** combining field-sensor ingestion, data-quality validation, and **water-balance scenario generation** for agricultural parcels.
- Orchestrated **6 specialized AI modules** across scientific watch, bibliography, project quality, decision support, and workflow coordination, using FastAPI, PostgreSQL/pgvector, and automated offline testing.

## EDUCATION

**National School of Advanced Sciences and Technologies of Borj Cedria (ENSTAB)**

*Engineering Degree in Advanced Technologies* — Specialization: Digitalisation & Data Analysis (DAD) Sep 2024 – Present

*Key courses: Generative AI, Advanced AI, Machine Learning, Image Processing, DevOps/MLOps, Big Data, Full-Stack Web Development, Databases, IoT, Cloud & Virtualization*

**Faculty of Sciences of Tunis (FST)**

*Preparatory Cycle: Mathematics and Physics*

National entrance exam for engineering schools 2024

## PROJECTS

**Syntern (Stagi) — AI Internship Simulator** • **2nd Place** • **TBS Atlas Hackathon**

- Built a multi-agent internship simulator with **4 role-specific AI agents** (Manager, Tech Lead, Client, Senior Intern), integrating real-time voice interaction through ElevenLabs and Web Speech API.
- Engineered an **n8n orchestration layer** for session management, agent routing, inactivity escalation, and behavioral evaluation, scoring users across **5 professional dimensions**.
- Developed a React/Vite workspace combining Slack-like communication, AI-generated Kanban tasks, Supabase persistence, and **automated GitHub repository code review**.

**GROOV — Multi-Agent AI Study Group**

- Built a local multi-agent learning platform with **6 pedagogical AI personas** powered by Ollama (LLaMA 3.2), with **4 adaptive study modes** and intelligent speaker orchestration.
- Engineered a **RAG pipeline** using FAISS and Sentence-Transformers for semantic PDF retrieval with page-level citations and document-grounded agent discussions.
- Developed multimodal learner-state adaptation by combining webcam-based emotion recognition with LLM sentiment analysis across **8 emotional states**, dynamically modifying teaching behavior.

- Implemented a Feynman-style evaluation workflow producing **0–100 feedback scores**, missing-concept detection, analogy assessment, and targeted improvement recommendations.

#### AgenticHire — AI-Powered Recruitment Platform — *PFA* • **3rd Place** • **NerdData ENSIT Hackathon**

- Architected a recruitment platform with **12 specialized AI agents** (7 candidate-side, 5 recruiter-side) orchestrated with Mistral AI and Hugging Face models.
- Built a multi-source job-ingestion pipeline aggregating listings from **11 platforms** through REST APIs, BeautifulSoup, and python-jobsby.
- Developed an end-to-end candidate intelligence workflow combining intent detection, CV analysis, semantic job matching, CV optimization, cover-letter generation, and recruiter-side candidate evaluation.
- Engineered the production-facing stack with a **FastAPI backend**, JWT authentication, SQLite/Pydantic data models, and a React/Vite conversational frontend.

#### Vendex — AI Forensic Claims Intelligence • **Top 6 of 37 Teams** • **GDGC FST Hackathon**

- Designed a **5-layer computer-vision forensic pipeline** for detecting manipulated, recycled, and AI-generated insurance-claim imagery.
- Combined **Error Level Analysis, FFT frequency analysis, DINOv2 embeddings, and Mahalanobis-distance anomaly detection** to identify visual inconsistencies across complementary forensic signals.
- Integrated **MobileSAM segmentation** for localized damage analysis and EXIF metadata forensics, served through asynchronous FastAPI inference.
- Developed a three-agent Mistral-based reasoning layer that synthesizes forensic evidence into an **explainable fraud assessment** and automated PDF report.

## SKILLS

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- **Programming:** Python, C++, Java, SQL, JavaScript, TypeScript
- **Machine Learning / Deep Learning:** PyTorch, Hugging Face Transformers, scikit-learn, OpenCV, DINOv2, MobileSAM, Anomaly Detection, Computer Vision, NLP
- **Generative AI / Agentic AI:** LLMs, Multi-Agent Systems, LangGraph, RAG, Mistral AI, Ollama, FAISS, Qdrant, Sentence-Transformers, n8n
- **AI Engineering:** FastAPI, REST APIs, Pydantic, PostgreSQL/pgvector, Supabase, Docker, Git, CI/CD, MLOps
- **Scientific Computing:** NumPy, SciPy, MATLAB, Signal Processing, Geospatial Data Processing
- **Frontend / Visualization:** React, TypeScript, Vite, TanStack Start, Three.js, react-three-fiber, Leaflet
- **Languages:** Arabic (Native), French (Fluent), English (Fluent)

## LEADERSHIP & ACTIVITIES

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- **ACM ENSTAB Club** Chairman — led club operations, technical training, competitive programming activities, and events | Sep 2025 – Sep 2026
- **ETC Junior Enterprise** — Project Management — coordinated project planning, team execution, client communication, and delivery tracking | Sep 2024 – Present
- **IEEE ENSTAB Student Branch** Member — technology events and collaborative engineering projects | Sep 2024 – Present
- **ElectroniX Club** Member — hardware/software engineering and IoT projects | Sep 2024 – Present

## CERTIFICATIONS

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#### Fundamentals of Deep Learning — *NVIDIA*

Issued: January 2025 | ID: 1t0OYs3ITkGJkXM3sgBKrQ

#### Certificate of Completion – Mathematics — *365 Data Science*

Issued: November 2024 | ID: CC-0427E007BE

#### AI Engineer for Data Scientists Certification — *DataCamp* ([In Progress](#))